SYSTEM ARCHITECTURE DOCUMENT

AI-Based Digital Twin Prototype for Two-Hour Glucose Forecasting

Happiest Health Digital Twin Challenge 2026

PROJECT INFORMATION

Challenge	Happiest Health Digital Twin Challenge 2026
Team name	Transcendental-Coders
Team leader	Madhav Kumar
Team members	1. Madhav Kumar 2. Sunny Kumar 3. Shubham Panchal
College	Kamla Nehru Institute of Technology
Incubator	N/A
Repository	https://github.com/shubhampanchal6376/digital-twin

PROJECT OVERVIEW

This project demonstrates an experimental workflow for processing Electronic Health Record (EHR) datasets and wearable glucose time-series data, and for forecasting glucose approximately **two hours (120 minutes)** ahead from historical wearable data. It is a proof of concept for experimental healthcare analytics.

The system includes:

1. EHR data inspection and preprocessing.
2. Wearable glucose data preprocessing.
3. Time-series feature engineering.
4. Machine-learning-based glucose forecasting.
5. A Flask API serving historical readings and predictions.
6. A web dashboard for visualizing historical glucose readings and forecast results.

120 min
Forecast horizon

5 min
Wearable sampling interval

18
EHR CSV files processed (recorded run, 0 failures)

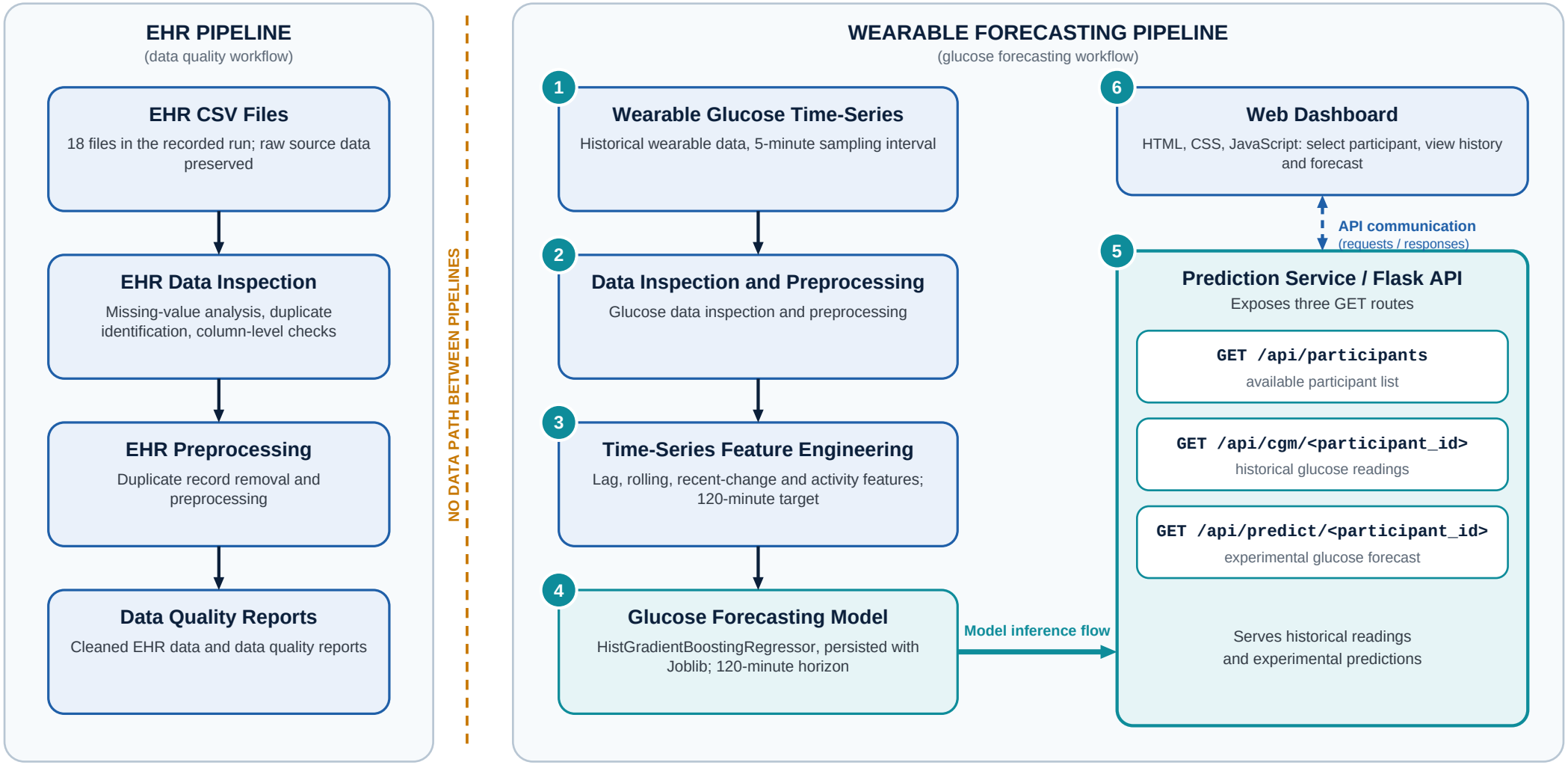
HistGradient-BoostingRegressor
Forecasting model

EXPERIMENTAL PROTOTYPE DISCLAIMER

This is an experimental healthcare analytics proof of concept. It has **not been clinically validated**, is based on **historical datasets**, and is not a real-time medical monitoring system. Its outputs do not constitute clinical decision support, and nothing in this document claims clinical effectiveness or guaranteed generalization. The EHR and wearable datasets are processed separately; individual-level mapping has not been established.

Complete System Architecture

Two separate pipelines within one project: EHR data quality pipeline and wearable glucose forecasting pipeline



EHR and wearable datasets are processed separately. Individual-level mapping has not been established.

The EHR pipeline does not feed the glucose forecasting model, and EHR information does not influence the current wearable glucose prediction.

LEGEND Data flow · Model inference flow · API communication · n Wearable pipeline stage order

Data Pipelines

Two independent preprocessing pipelines: EHR data quality and wearable glucose forecasting inputs

A. EHR DATA PIPELINE

EHR CSV Files
18 files, recorded run

→

EHR Data Inspection
missing values, duplicates, columns

→

EHR Preprocessing
duplicate record removal

→

Data Quality Reports
cleaned data and reports

Input and processing

- **Input:** EHR CSV files.
- Data inspection and missing-value analysis.
- Duplicate record identification and removal.
- Column-level data quality checks.
- Preprocessing and quality report generation.

Output and recorded result

- **Output:** cleaned EHR data and data quality reports.
- **Recorded result:** 18 EHR CSV files processed with zero processing failures in the recorded run.
- The raw source data was preserved.
- This pipeline is separate from the wearable forecasting pipeline.

B. WEARABLE GLUCOSE FORECASTING PIPELINE

Wearable Glucose Data
5-minute sampling

→

Inspection, Preprocessing
glucose data checks

→

Feature Engineering
lag, rolling, activity

→

Target Construction
24-step shift = 120 min

→

Forecasting Model
HistGradientBoosting

MODEL INPUT FEATURES (WEARABLE DATA)

Historical lag features

Earlier glucose readings are supplied as lagged features so the model sees recent history.

Rolling glucose statistics

Rolling statistics summarize glucose behaviour over recent windows of readings.

Recent glucose change

Features describing how glucose has recently changed between readings.

Activity features

Available wearable activity features such as steps and calories.

Two-hour target construction

Sampling interval: 5 minutes. Forecast horizon: 120 minutes. A **24-step shift** corresponds to 24 x 5 min = **120 minutes**, so the target is the glucose value approximately two hours after each time step. **The future target is the prediction label only and is never used as an input feature.** Feature engineering is applied to the wearable data only; no EHR variables are used.

Model, API and Dashboard

Recorded experimental results for the glucose forecasting prototype

FORECASTING MODEL

HistGradientBoostingRegressor (Scikit-learn) predicts glucose approximately two hours (120 minutes) ahead using historical glucose-related features and available wearable activity features. The trained model is persisted with Joblib. It is compared against a **persistence baseline**, which carries the current glucose value forward as the forecast.

RECORDED EVALUATION RESULTS

Model	MAE	RMSE	R^2
Persistence baseline	33.474	46.229	0.1376
HistGradientBoostingRegressor	**30.234**	**39.879**	**0.3583**

Table 1. Recorded experimental evaluation results. Lower MAE and RMSE and higher R^2 are better. Measurement units are not documented for these metrics and are therefore not stated.

Relative MAE difference

Recorded ML model MAE is **approximately 9.7% lower** than the persistence baseline MAE.

Interpretation

In the recorded evaluation the ML model shows lower MAE and RMSE and a higher R^2 than the baseline. An R^2 of 0.3583 still leaves much of the variation unexplained. These are experimental results only; they do not establish clinical effectiveness, clinical validation, or guaranteed generalization.

FLASK API ENDPOINTS

Endpoint	Returns
`GET /api/participants`	Available participant list
`GET /api/cgm/<participant_id>`	Historical glucose readings
`GET /api/predict/<participant_id>`	Experimental glucose forecast

WEB DASHBOARD RESPONSIBILITIES

- Lets the user select a wearable participant.
- Displays historical glucose readings.
- Displays experimental forecast results.
- Communicates with the Flask API for all data.

RECORDED PREDICTION EXAMPLE

Participant	`HUPA0001P`
Current glucose	`326.0`
Predicted glucose after two hours	`294.2235`
Forecast horizon	`120 minutes`
Latest source timestamp	`2018-06-27T23:55:00`
Dataset status	`historical_dataset`
Model status	`experimental_prototype`

Recorded historical-dataset response, shown as an example only. It is not a real-time or clinical prediction and does not imply a medically meaningful result. Glucose units are not stated.

Limitations and Conclusion

Scope boundaries of the current experimental prototype

1 EHR and wearable data are not linked

The datasets are processed through separate pipelines. Individual-level mapping has not been established, and EHR information does not influence the current wearable glucose model.

2 Not clinically validated

The prototype has not been clinically validated and must not be used to guide clinical decisions.

3 Historical dataset

Readings and predictions come from historical data. The system is not a real-time monitoring tool.

4 Metrics do not show clinical usefulness

Recorded MAE, RMSE and R^2 compare the model with a persistence baseline only. Model performance does not establish clinical usefulness or guaranteed generalization.

5 Additional validation required

Further evaluation is needed before any broader claims can be made about the model's behaviour.

6 Not clinical decision support

The dashboard visualizes historical readings and experimental forecasts only.

Future work

Possible next steps include data reconciliation between the EHR and wearable datasets, if participant-level linkage can be established, and further participant-level evaluation of forecasting performance.

PROJECT SUMMARY

Transcendental-Coders (Team leader: Madhav Kumar; members: Madhav Kumar, Sunny Kumar, Shubham Panchal; Kamla Nehru Institute of Technology) present an experimental proof of concept for the Happiest Health Digital Twin Challenge 2026. The project combines an EHR data quality pipeline with a separate wearable pipeline that uses a HistGradientBoostingRegressor to forecast glucose approximately two hours ahead from historical data. A Flask API (`/api/participants`, `/api/cgm/<participant_id>`, `/api/predict/<participant_id>`) feeds a web dashboard that visualizes historical readings and forecasts. In the recorded evaluation, the model's MAE was approximately 9.7% lower than a persistence baseline. These results are experimental, based on a historical dataset, and not clinically validated.

Repository: https://github.com/shubhampanchal6376/digital-twin